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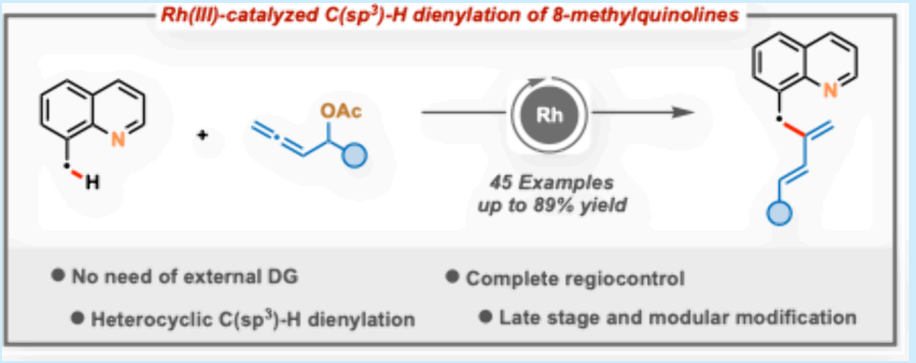
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Access to Branched 1,3-Dienes *via* Innate C(sp^3)−H Functionalization of 8-Methylquinolines

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ABSTRACT: Allenes have gained prominence in transition metal-catalyzed C−H activation reactions due to their versatile reactivity. While C(sp^2)−H functionalization with allenes has been well explored, the functionalization of more challenging C(sp^3)−H bonds with allenes remains largely underexplored. Herein, we present a scalable Rh(III)-catalyzed strategy for the site-selective C(sp^3)−H dienylation of biologically relevant 8-methylquinolines employing allenyl carbinol acetates for accessing structurally diverse 1,3-dienes. Leveraging the innate directing group ability of the quinoline nitrogen, our protocol promotes the selective 2,1-migratory insertion of allenes. The transformation exhibits a broad substrate scope and enables the late-stage construction of densely functionalized 1,3-diene conjugates with biologically relevant motifs, highlighting the synthetic utility and broad applicability of this method.

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Conjugated dienes are ubiquitous scaffolds widely present in biologically active natural products and serve as key synthetic precursors in the preparation of functional polymers and complex molecular frameworks (Scheme 1A).[1] Also, their distinctive reactivity makes them valuable substrates in a variety of organic transformations, including 1,4-functionalizations, cycloadditions, and other pericyclic reactions.[2] Conventional approaches for diene synthesis such as Wittig olefination, enyne metathesis, and transition metal-catalyzed cross-coupling reactions rely on prefunctionalized substrates such as carbonyl compounds, halides, or boronic acids, limiting modularity and substrate diversity.[3,4] Consequently, strategies enabling modular access to structurally diverse conjugated dienes from simple substrates are highly desirable. In this context, catalytic dienylations leveraging reactive π-systems as dienylating agents have recently been investigated.[5]

On the other hand, allenes have garnered particular attention due to their unique structure and versatile reactivity.[6] In particular, allenyl carbinol derivatives bearing a heteroatom at the α-position are found to be useful substrates in transition metal catalysis.[7] The heteroatom guides the insertion pathway by coordinating with the metal catalyst and facilitates the β-elimination. In parallel, transition metal-catalyzed C−H functionalization has emerged as a powerful approach in organic synthesis, allowing the direct transformation of otherwise inert C−H bonds into valuable functionalities.[8] Integration of this prevailing strategy with allene chemistry offers a modular route for accessing conjugated dienes from simple substrates (Scheme 1B). A pioneering report by Glorius and co-workers demonstrated Rh(III)-catalyzed C(sp^2)−H dienylation of benzamides employing allenyl carbinol carbonates.[9] Building on this, our group[10] and the Ma group[11] explored Co(III)-catalyzed C8 dienylation of quinoline *N*-oxides and Rh(III)-catalyzed stereoselective C2 dienylation of indoles, respectively. All three protocols involve the selective formation of M-σ-allyl species **Int-I** via regioselective 2,1-migratory insertion. In sharp contrast, recently Cembellín and co-workers demonstrated a mechanistically distinct 3,2-

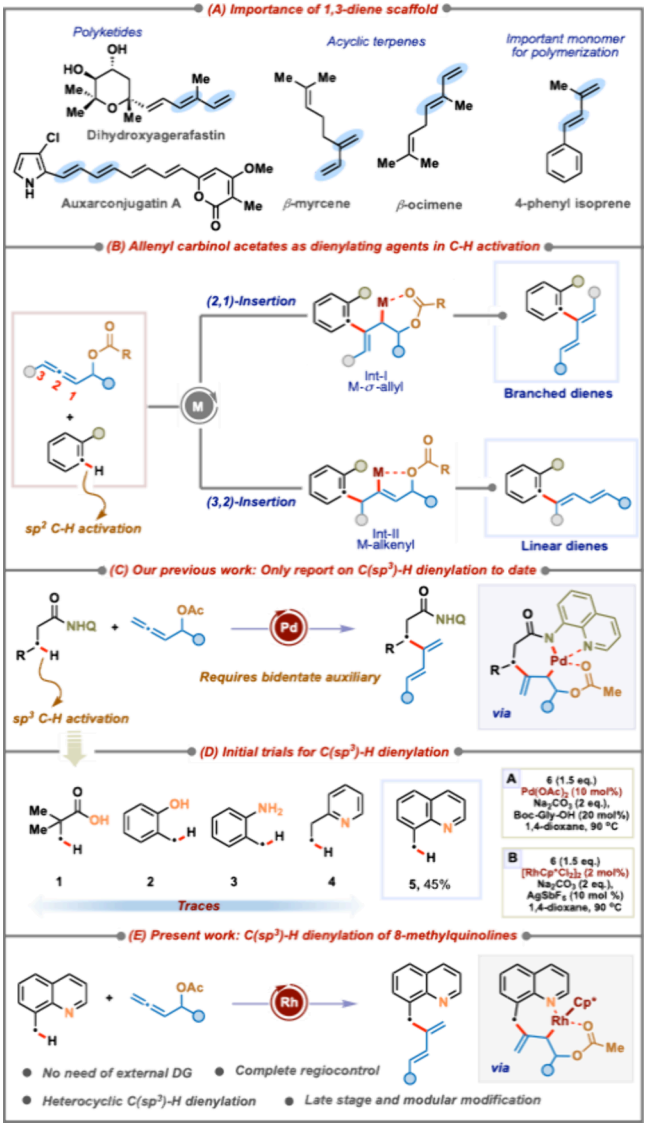
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Scheme 1. Overview



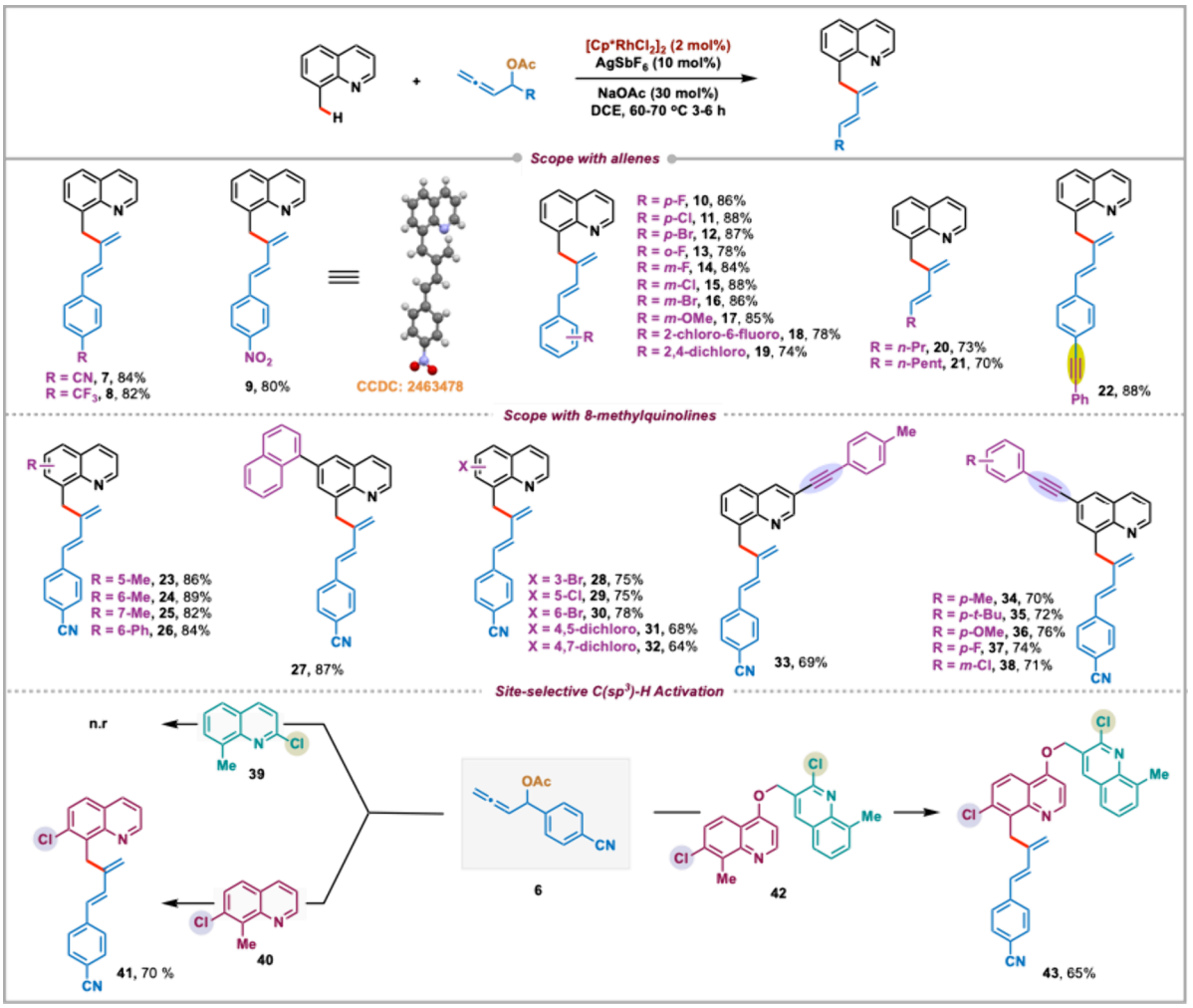
migratory insertion, affording linear dienes through M-alkenyl intermediate **Int-II** under Mn catalysis (Scheme 1B).[12] Despite these advances, C−H dienylation with allenes remains largely confined to C(sp^2)−H bonds. Extension of this transformation to aliphatic C(sp^3)−H bonds is particularly appealing but remains challenging due to the intrinsically lower reactivity, the lack of π-coordination, and the weaker overlap of C(sp^3)−H bonds with metal centers.[13] Our group addressed this gap by reporting the first example of β- and γ-C(sp^3)−H dienylation of aliphatic amides using allenyl carbinol acetates to access sp^3-functionalized conjugated dienes (Scheme 1C).[14] However, this strategy still relied on an 8-aminoquinoline directing group and finely tuned ligands, limiting its practicality. To overcome these constraints, we sought a more straightforward C(sp^3)−H dienylation protocol leveraging the inherent directing capabilities of native functional groups. The key challenge is the weak coordination ability of these groups, which hampers metallacycle formation that is critical for effective C−H activation and allene insertion. Toward this end, we evaluated a series of substrates bearing native functionalities with allenyl carbinol acetate **6** under Pd(II) and Rh(III) catalysis (Scheme 1D). Substrates containing weakly coordinating groups such as pivalic acid **1**, *o*-methylphenol **2**, and *o*-methylaniline **3** failed to show any reactivity. 2-Ethylpyridine **4** having a strongly coordinating nitrogen donor is also ineffective, likely due to the conformational flexibility hindering stable metallacycle formation. These cumulative findings pointed toward the necessity of a more rigid system capable of fostering stable metal−substrate interactions. We then turned toward 8-methylquinoline **5**, a structurally constrained system featuring a benzylic C(sp^3)−H bond amenable to C−H activation and a nitrogen atom capable of coordinating to a metal catalyst.[15] This built-in coordination could facilitate cyclometalation resulting in five-membered metallacycle formation. Moreover, quinoline-based motifs are highly prevalent in biologically active molecules and pharmaceutical agents, thereby rendering them highly valuable targets for selective functionalization and late-stage molecular diversification.[16] To our delight, 8-methylquinoline **5** was found to be an amenable substrate, affording the desired product **7** in 45% yield when treated with allene **6** and 2 mol % $[RhCp^*Cl_2]_2$ (condition B). Encouraged by this preliminary result, we undertook a comprehensive optimization study, which ultimately identified a rhodium-based catalytic system as optimal, enabling the efficient and selective C(sp^3)−H dienylation of 8-methylquinoline derivatives (Scheme 1E).

After the encouraging observation, various reaction parameters such as the solvent, additive, base, and catalyst were extensively optimized employing 8-methylquinoline **5** and allene **6** to identify the optimal conditions (see the Supporting Information for optimization details). The desired product **7** was isolated in 84% yield under the optimized reaction conditions: 2 mol % $[Cp^*Rh(III)Cl_2]_2$, 10 mol % $AgSbF_6$, and 30 mol % NaOAc in DCE at 65 °C (see the Supporting Information for the optimization details). While solvents such as DCM and TFE resulted in slightly inferior yields, other solvents such as MeCN and THF led to the formation of dienylated product **7** in lower yields. The choice of base was also crucial, as replacing NaOAc with either AgOAc or NaOPiv led to reduced conversions. In addition, the use of additives like AgOTf and $AgBF_4$ could not enhance the yield, suggesting $AgSbF_6$ is the best additive. Notably, the yield of **7** decreased considerably when the reaction was carried out in the absence of NaOAc or $AgSbF_6$. Other metal catalysts such as $Pd(OAc)_2$ and $[Cp^*Co(CO)I_2]$ were found to be ineffective, highlighting the effectiveness of Rh(III) for the transformation.

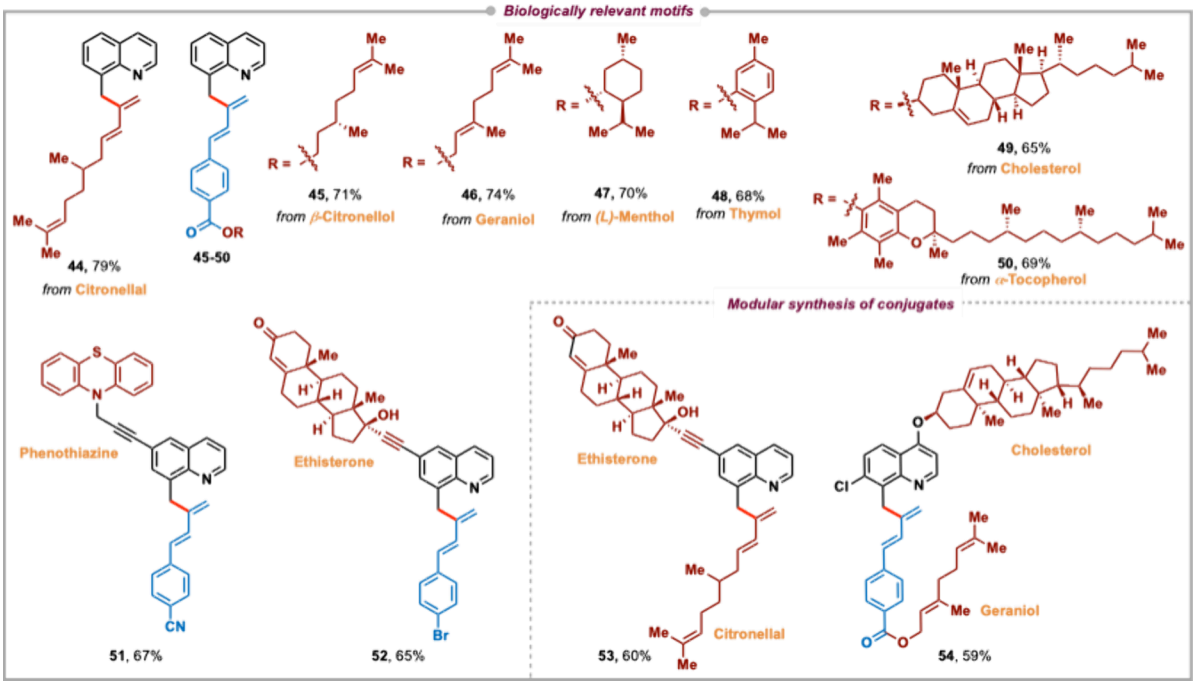
After establishing the optimized reaction conditions, we proceeded to investigate the generality of the developed protocol with diversely substituted allenyl carbinol acetates employing 8-methylquinoline **5** (Scheme 2). Allenes having electron-withdrawing groups or halogens at the *para* position of the phenyl ring afforded the desired products **7**−**12** in good to excellent yields (80−88%). However, the *o*-fluoro derivative showed slightly reduced efficiency, delivering product **13** in 78% yield. Single-crystal X-ray diffraction analysis of nitro compound **9** unambiguously confirmed the structure and stereochemistry of the diene. Next, we tested the presence of halogens and the methoxy group at the *meta* position of the phenyl ring, and these were also found to be compatible under the reaction conditions to furnish **14**−**17** in good yields (84−88%). Furthermore, 2,6- and 2,4-dihalosubstituted substrates were well tolerated, delivering products **18** and **19** in moderate to good yields. Notably, allenyl carbinol acetates bearing aliphatic groups were also found to be amenable under our conditions to provide **20** and **21** in moderate yields (70−73%).

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Scheme 2. Substrate Scope of Dienylation of 8-Methylquinolines

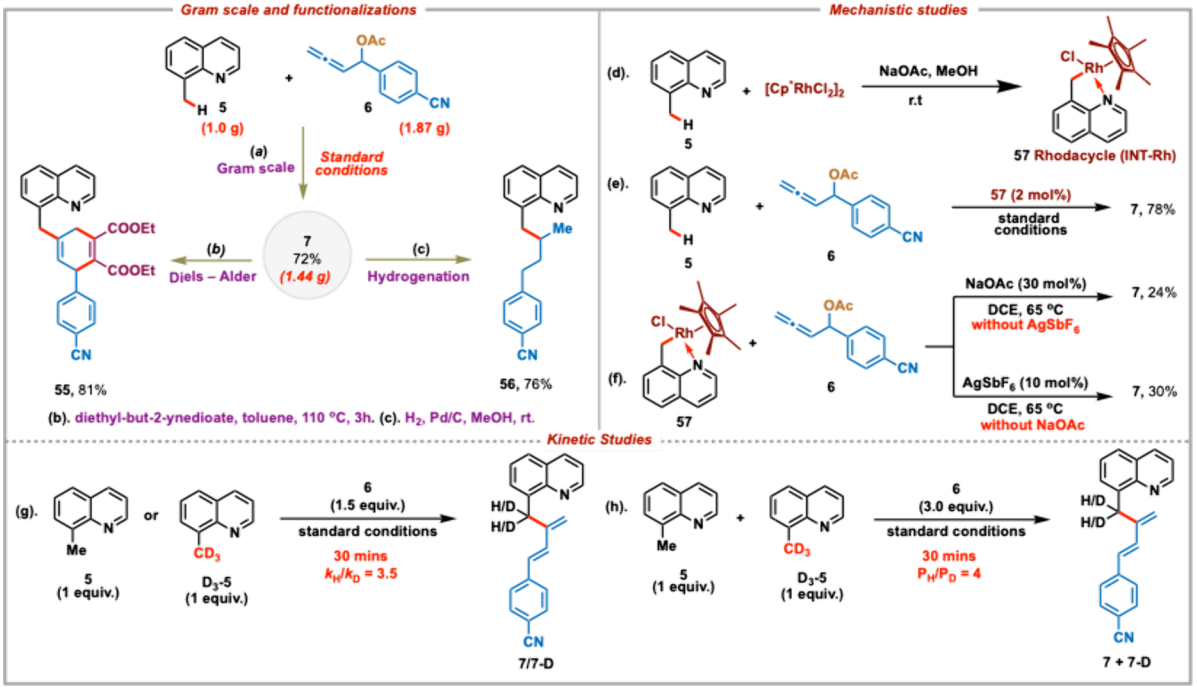


Scheme 3. Substrate Scope of Natural Product-Tethered Dienylated Quinoline Scaffolds



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Scheme 4. Gram-Scale, Functionalization, and Mechanistic Studies



It is worth mentioning that allene having an alkynyl functionality reacted smoothly as a dienylating agent (**22**, 88%), keeping the alkyne functionality intact showing the superior reactivity of allene over alkyne.

We then embarked on exploring the scope of 8-methylquinolines under the optimized conditions employing allene **6** to synthesize functionalized 8-methylquinolines featuring the 1,3-diene motif. Quinolines having a methyl group at various positions yielded the desired dienylation products **23**−**25** in good to excellent yields ranging from 82% to 89%. Both phenyl and naphthyl groups at the C6 position of 8-methylquinoline were found to be compatible to furnish dienylated products **26** and **27**, respectively, in 84−87% yields. However, the use of Br- or Cl-substituted 8-methylquinolines led to a slight decrease in the yield of products **28**−**32** (64−78%). A variety of C3- and C6-alkynylated quinoline derivatives **33**−**38** having electron-donating or electron-withdrawing groups on the alkynyl system were synthesized in 69−76% yields. Unfortunately, substitution at the C2 position (with methyl or chloro) completely suppressed reactivity, which might be due to impeding Rh coordination with the quinoline nitrogen. Interestingly, treating 7-chloro-8-methylquinoline **40** with **6** provided the desired product **41** in 70% yield. With this observation, we envisioned that this substrate-dependent reactivity could be exploited in a sterically selective manner and tested bisquinoline substrate **42** having two different potential sites for the reaction. Pleasingly, when **42** was subjected to the optimized reaction conditions, monodienylated product **43** was obtained selectively in 65% yield with the 8-methylquinoline scaffold having the 2-Cl substitution unperturbed.

To demonstrate the applicability of our developed protocol, allenes tethered with biologically relevant and naturally occurring alcohol derivatives such as citronellal, β-citronellol, geraniol, menthol, thymol, cholesterol, and tocopherol were tested. Conjugated dienes **44**−**50** were obtained in good yields of up to 79% (Scheme 3). Owing to the appealing biological properties of phenothiazine and ethisterone, 8-methylquinolines tethered with these entities were prepared and subjected to dienylation with allenyl carbinol acetates to furnish conjugates **51** and **52** in 67% and 65% yields, respectively. Application of our Rh(III)-catalyzed dienylation protocol for the modular synthesis of conjugates was then assessed by subjecting 8-methylquinoline tethered with ethisterone and citronellal-derived allenyl carbinol acetate under the optimized conditions to isolate complex heterocyclic scaffold **53** in 60% yield. Similarly, cholesterol-conjugated 8-methylquinoline was coupled with geraniol-derived allene to synthesize **54** in 59% yield.

To illustrate the practicality of the present reaction protocol, a gram-scale reaction was conducted with 1.0 g of 8-methylquinoline **5** and 1.87 g of allene **6** to isolate 1.44 g of dienylation product **7** (72% yield) (Scheme 4a). To gain insights into the reaction mechanism, rhodacyclic complex **57** was synthesized via the cyclometalation of 8-methylquinoline **5** at room temperature in MeOH, which was then employed as a catalyst precursor (2 mol %) in place of $[Cp^*RhCl_2]_2$ under the standard reaction conditions (Scheme 4d,e). The reaction resulted in the formation of **7** in 78% yield, highlighting its catalytic efficiency and significance in C−H activation. A stoichiometric reaction using complex **57** and allene **6** in the absence of $AgSbF_6$ produced product **7** in a diminished yield of 24% (Scheme 4f). Similarly, the reaction led to a poor yield of 30% in the absence of NaOAc, highlighting the importance of a cationic rhodacycle for allene coordination and carborhodation. Additionally, kinetic isotope effect (KIE) studies were performed with **5** and 8-(methyl-d_3) quinoline **D_3-5** (Scheme 4g). Parallel KIE experiments with **5** and **D_3-5** using **6** gave a value of 3.5, and the competitive experiment between **5** and **D_3-5** yielded a product distribution value of 4 (Scheme 4h). Both of these experimental KIE values suggest that the C−H bond cleavage is likely to be involved in the rate-determining step. Based on these preliminary studies, the plausible

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mechanism for the dienylation reaction is proposed (see the Supporting Information).

In summary, we have developed Rh(III)-catalyzed dienylation of 8-methylquinolines employing allenyl carbinol acetates via challenging C(sp^3)−H bond activation in the absence of any exogenous directing group and ligand. The innovative use of these reactive allenes opened a new avenue for expanding the scope of more challenging C(sp^3)−H bond activation, and a variety of 8-methylquinoline derivatives were efficiently functionalized to afford diversely functionalized 1,3-dienes. The value of the developed catalytic approach was exemplified by successful modification of biologically active scaffolds. Furthermore, the utility of the developed methodology was demonstrated through its seamless scalability and various synthetic transformations of the product.

■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and its Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.5c02926.

Additional experimental procedures and spectroscopic data for the synthesized compounds (PDF)

Accession Codes

Deposition Number 2463478 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge via the joint Cambridge Crystallographic Data Centre (CCDC) and Fachinformationszentrum Karlsruhe Access Structures service.

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Author Contributions

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Notes

The authors declare no competing financial interest.

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